

Testing Results for Different Components

Component	Probe D	Probe C	Red LED	Green LED	Result
Diode	1st terminal 2nd terminal	2nd terminal 1st terminal	Off On	On Off	Probe D side is anode (p) and probe C side is cathode (n) Probe D side is 'n' and probe C side is 'p'
Transistors	C	E	X	X	Unused pin is base
Any type pnp or npn	E	C	X	X	Unused pin is base
nnp-type transistor	B	E	On	Off	p-n junction
	B	C	On	Off	p-n junction Result: 'p' is common, so npn transistor
pnp-type transistor	B	E	Off	On	n-p junction
	B	C	Off	On	n-p junction Result: 'n' is common, so pnp transistor
Step-down transformer	Primary terminal 1	Primary terminal 2	Glow with low intensity	Glow with low intensity	Both LEDs glowing with low intensity Result: Primary side
	Secondary terminal 1	Secondary terminal 2	Glow with high intensity	Glow with high intensity	Both LEDs glowing with high intensity Result: Secondary side
Continuity	Connect with LEDs probe	X	On	Off	Indicates shorting

inserted into it during testing.) Slightly unequal probe lengths help to make easy contacts.

Assemble the circuit on a general-purpose PCB which is as compact as

possible and put it inside a glue stick tube (whose inner mechanism has been removed) at its centre. The metallic disk and metallic strips can be cut out from any tin container. For the probes,

use the spring mechanism of gel ball pens. Probes C and D are the points representing the probe terminals. Two button cells (CR2032) are used to power the probe circuit. ●

ROOM SOUND MONITOR

■ PRADEEP G.

With this simple circuit, you can secretly listen to conversations going on in a room. The circuit is very sensitive and powered by a 3V battery.

The working of the circuit is simple. When a conversation is going on in the vicinity of the condenser

microphone, the weak sound signals are amplified by transistors T1 and T2. The amplified signal is fed to audio driver transformer X1, which further amplifies the signal. The transformer output is fed to a speaker through a capacitor. In brief, when the microphone senses sound in a room, the same can be listened outside the room through the speaker. Keep the speaker away

from the microphone to avoid howling due to feedback. Transformer X1 is a readily available six-leaded driver transformer having a single primary winding and dual secondary windings. Adjust preset VR1 for maximum clarity and gain.

Assemble the circuit on a general-purpose PCB and house in a suitable cabinet. Connect the condenser microphone through a wire and place the cabinet anywhere in the target room. Connect output wires to a 0.5-watt speaker kept in the room where you want to monitor from. Any type of two-core wire can be used for the purpose. The circuit can pick up sound signals from a distance of several metres. Use a good-quality microphone. It can be hidden inside a 2-3cm long tube. ●

